

# Healthcare Clinical Intelligence Platform (HCIP)

## Product Requirements Document (PRD)

### Version

v1.0

### Status

Planning

### Owner

Mohan Rao

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## 1. Product Vision

Build a production-grade, multi-tenant Healthcare Clinical Intelligence Platform that enables healthcare professionals to retrieve, analyze, verify, and interact with medical knowledge using Agentic RAG.

The platform must:

- Support enterprise healthcare organizations
  - Serve 10,000+ queries/day
  - Provide evidence-based responses
  - Minimize hallucinations
  - Maintain complete auditability
  - Support regulatory compliance requirements
  - Offer full observability and evaluation
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## 2. Target Users

### Primary Users

#### Doctor

Uses system for:

- Clinical decision support

- Treatment recommendations
- Research assistance
- Guideline lookup

### **Pharmacist**

Uses system for:

- Drug interaction analysis
- Dosage validation
- Contraindication review

### **Researcher**

Uses system for:

- Literature review
  - Medical evidence synthesis
  - Clinical research support
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## **Secondary Users**

### **Hospital Administrator**

Uses system for:

- Knowledge management
  - User management
  - Usage monitoring
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## **3. Core Product Goals**

### **Goal 1**

Provide evidence-based healthcare knowledge retrieval.

### **Goal 2**

Reduce medical knowledge search time.

### **Goal 3**

Enable safe AI-assisted medical reasoning.

#### **Goal 4**

Support enterprise-scale healthcare organizations.

#### **Goal 5**

Deliver explainable and auditable AI responses.

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## **4. Core Use Cases**

### **UC-01 Clinical Question Answering**

Example:

Can diabetic patients taking Metformin safely receive contrast CT imaging?

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### **UC-02 Drug Interaction Analysis**

Example:

Check interaction between Metformin and Ciprofloxacin.

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### **UC-03 Hospital SOP Search**

Example:

Show sepsis treatment protocol.

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### **UC-04 Medical Research Assistant**

Example:

Summarize latest evidence on hypertension treatment.

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### **UC-05 Guideline Comparison**

Example:

Compare WHO vs CDC recommendations.

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## 5. Multi-Tenant SaaS Requirements

Each healthcare organization must have:

- Isolated users
- Isolated documents
- Isolated conversations
- Isolated permissions
- Isolated analytics

Tenant data must never leak between organizations.

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## 6. Supported Data Sources

### Internal Sources

- SOPs
- Clinical Guidelines
- Treatment Protocols
- Drug Formularies
- Insurance Policies
- Patient Education Material

### External Sources

- PubMed
  - WHO
  - FDA
  - CDC
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## 7. Supported File Types

### Documents

- PDF
- DOCX
- PPTX
- XLSX

- CSV
- TXT

## Images

- PNG
- JPG
- TIFF

## Structured Data

- JSON
  - XML
  - FHIR
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# 8. Ingestion Requirements

The platform must:

- Parse documents
  - Extract metadata
  - Extract tables
  - Extract images
  - Extract charts
  - Perform OCR
  - Generate embeddings
  - Create knowledge graph entities
  - Apply RBAC metadata
  - Maintain document versions
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# 9. Retrieval Requirements

The platform must support:

- Semantic Search
- Hybrid Search
- BM25 Retrieval
- Metadata Filtering
- RBAC Filtering
- Graph Retrieval
- Reranking
- Context Compression

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## 10. Agent Architecture

### Planner Agent

Responsibilities:

- Query classification
- Retrieval strategy selection
- Source selection

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### Retriever Agent

Responsibilities:

- Execute retrieval
- Aggregate context
- Return evidence

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### Verifier Agent

Responsibilities:

- Citation validation
- Contradiction detection
- Hallucination checks

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### Safety Agent

Responsibilities:

- High-risk medical detection
  - Emergency detection
  - Escalation recommendations
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## Response Agent

Responsibilities:

- Final answer generation
  - Citation generation
  - Confidence scoring
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## 11. User Roles

### Super Admin

Platform-level management

### Organization Admin

Organization management

### Doctor

Clinical usage

### Pharmacist

Drug-related usage

### Researcher

Research usage

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## 12. Functional Requirements

### Authentication

- Login
- Logout
- Password Reset

### User Management

- Invite Users

- Role Assignment
- User Deactivation

## **Chat**

- Multi-session chat
- Chat history
- Source citations

## **Documents**

- Upload
- Delete
- Reindex
- Versioning

## **Analytics**

- Usage statistics
- Query statistics
- Cost monitoring

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# **13. Evaluation Requirements**

The system must evaluate:

- Faithfulness
- Context Recall
- Context Precision
- Answer Relevancy
- Hallucination Rate
- Citation Coverage

Tools:

- RAGAS
- DeepEval

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# **14. Observability Requirements**

Track:

- Agent execution
- Retrieval latency



- LLM latency
- Token usage
- Cost per query
- Retrieval quality
- User feedback

Tools:

- Langfuse
  - OpenTelemetry
  - Grafana
  - Prometheus
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## 15. Security Requirements

- RBAC
  - JWT Authentication
  - Encryption at Rest
  - Encryption in Transit
  - Audit Logging
  - PII Masking
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## 16. Compliance Requirements

Architecture should be compatible with:

- HIPAA
  - GDPR
  - SOC2
  - ISO 27001
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## 17. Performance Requirements

Target:

- 10,000+ Queries/Day
  - <5 Second Average Response Time
  - 99.9% Availability
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## 18. MVP Scope

Included:

- Multi-Tenant SaaS
- Document Upload
- Agentic RAG
- PubMed Integration
- Hybrid Search
- Citations
- Confidence Scores
- Observability
- Evaluation

Excluded:

- Fine-tuned Models
  - EHR Integration
  - Voice Interface
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## 19. Success Metrics

- Retrieval Precision > 85%
  - Hallucination Rate < 5%
  - Average Response Time < 5s
  - Citation Coverage > 95%
  - User Satisfaction > 4.5/5
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## 20. Future Roadmap

### V2

- EHR Integration
- FHIR APIs
- Voice Assistant
- Multimodal Clinical Reports

### V3

- Predictive Clinical Insights
- Clinical Workflow Automation
- Advanced Knowledge Graph Reasoning